

1.6 Electrochemistry

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.6 Electrochemistry
Difficulty	Medium

Time allowed: 20

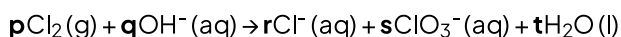
Score: /9

Percentage: /100

Question 1

Oxidation numbers can be used to balance equations. Chlorine and hot aqueous sodium hydroxide react to produce chloride ions, chlorate ions and water.

What are the values of the coefficients p, r and s in the equation?



	p	r	s
A	3	5	1
B	3	6	2
C	2	5	1
D	2	4	2

[1 mark]

Question 2

The chemistry of the Group VII elements often involves redox processes.

Which statement is correct?

- A. Bromine can oxidise chloride ions.
- B. Iodide ions are the weakest reducing agent of the first four Group VII ions.
- C. In reactions with water, chlorine is oxidised and reduced.
- D. Fluorine is a weaker oxidising agent than chlorine.

[1 mark]

Question 3

Four reactions are shown. In which reaction is the species in bold acting as an oxidising agent?

- A. $\text{Cr}_2\text{O}_7^{2-} + 8\text{H}^+ + \mathbf{3\text{SO}_3^{2-}} \rightarrow 2\text{Cr}^{3+} + 4\text{H}_2\text{O} + 3\text{SO}_4^{2-}$
- B. $\text{Mg} + \mathbf{\text{Fe}^{2+}} \rightarrow \text{Mg}^{2+} + \text{Fe}$
- C. $\text{Cl}_2 + \mathbf{2\text{Br}^-} \rightarrow 2\text{Cl}^- + \text{Br}_2$
- D. $\text{Fe}_2\text{O}_3 + \mathbf{3\text{CO}} \rightarrow 2\text{Fe} + 3\text{CO}_2$

[1 mark]

Question 4

In which compound are there two different elements with the same oxidation number?

- A. $\text{Mg}(\text{OH})_2$
- B. Na_2SO_4
- C. HClO
- D. NH_4Cl

[1 mark]

Question 5

When sulfuric acid and sodium iodide react one of the reactions that take place is shown by the equation below



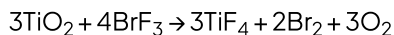
Which species has been oxidised in this reaction?

- A. I^-
- B. SO_4^{2-}
- C. Na^+
- D. H^+

[1 mark]

Question 6

The following reaction can be used to determine the mass of titanium dioxide in an ore.



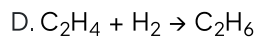
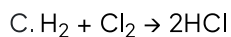
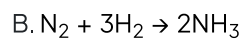
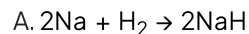
Which element increases in oxidation number in this reaction?

- A. Fluorine
- B. Bromine
- C. Titanium
- D. Oxygen

[1 mark]

Question 7

In which reaction does hydrogen behave as an oxidising agent?

**[1 mark]****Question 8**

Chlorine gas reacts with both hot and cold aqueous sodium hydroxide, each resulting in a different set of products.

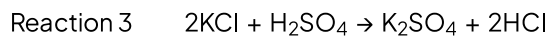
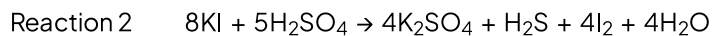
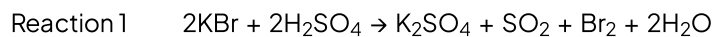
What are the oxidation numbers of chlorine in the products of these reactions?

	Hot aqueous sodium hydroxide	Cold aqueous sodium hydroxide
A	-1, +5	-1, +1
B	-1, +5	-1, +2
C	+1, +6	-1, +1
D	+1, +6	-1, +2

[1 mark]

Question 9

When solid potassium halides are added to concentrated sulfuric acid, the following reactions take place:



In each reaction, what is the largest change in the oxidation number of sulfur?

	Reaction 1	Reaction 2	Reaction 3
A	1	4	1
B	2	4	0
C	2	8	0
D	4	8	1

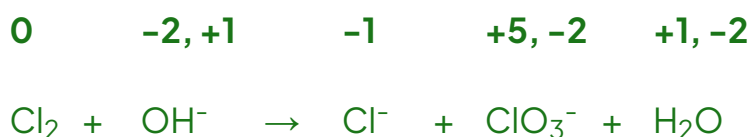
[1 mark]

Model Answers: Medium

1

The correct answer is **A** because:

- Write oxidation numbers of each species in the unbalanced equation.



- Write down the changes in oxidation number:
 - First Cl atom (from Cl_2): 0 to -1 so change is -1.
 - Second Cl atom: 0 to +5 so change is +5.
- Multiply first Cl atom by 5 to balance the changes in oxidation number.
 - This gives a total of 6 Cl (or 3Cl_2), 5 that decrease by one oxidation number unit and one that increases by 5 oxidation number units.
- Balance the O and H by adding 6 OH^- on the left and 3 H_2O on the right.
- Check the charges balance and write the full, balanced equation:
 - $3\text{Cl}_2 + 6\text{OH}^- \rightarrow 5\text{Cl}^- + \text{ClO}_3^- + 3\text{H}_2\text{O}$
- The coefficient of p is thus 3, of r is 5 and of s is 1.

B, C & D are incorrect as these options do not correctly balance the equation.

2

The correct answer is **C** because:

- Chlorine dissolves in water to produce hydrochloric acid and hypochlorous acid.
 - $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HOCl}$
- If we analyse the half reactions, we can see that this is a **disproportionation**

reaction in which chlorine is both oxidised and reduced.

- Oxidation:
 - $\frac{1}{2}\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HOCl} + \text{H}^+ + \text{e}^-$ (change in oxidation number Cl: 0 to +1).
- Reduction:
 - $\frac{1}{2}\text{Cl}_2 + \text{e}^- \rightarrow \text{Cl}^-$ (change in oxidation number Cl: 0 to -1).

A & D are incorrect as the oxidising power in Group VII **increases as you move up**, with fluorine being the most powerful oxidising agent.

B is incorrect as the reducing power of the halide ions **increases as you move down** due to increasing radius and shielding effects. As iodine is lower than F, Cl and Br, its ions are more powerful reducing agents.

3

The correct answer is **B** because:

- Oxidising agents are substances that oxidise other species, gain electrons and are themselves reduced.
- **Write down the oxidation numbers of each species in the reaction.**



- In equation **B**, Fe^{2+} oxidises $\text{Mg}(0)$ to $\text{Mg}^{2+}(+2)$ and is itself reduced from $\text{Fe}^{2+}(+2)$ to $\text{Fe}(0)$.

A, C & D are incorrect as in these reactions, each of the species in bold undergo oxidation so they cannot be acting as oxidising agents.

4

The correct answer is **C** because:

- The oxidation number of oxygen is -2; hydrogen is +1 and chlorine is also +1 so

hydrogen and chlorine have the same oxidation number.

A is incorrect as in $\text{Mg}(\text{OH})_2$ the oxidation number of magnesium is +2, oxygen is -2 and hydrogen is +1.

B is incorrect as in Na_2SO_4 the oxidation number of sodium is +1, sulfur is +6 and oxygen is -2.

D is incorrect as in NH_4Cl the oxidation number of nitrogen is -3, hydrogen is +1 and chlorine is -1.

5

The correct answer is **A** because:

- Oxidation refers to the **loss of electrons** and reduction refers to the **gain of electrons**: OIL-RIG (oxidation is loss - reduction is gain).
- In the compound NaI , sodium has an oxidation state of +1, and iodine an oxidation state of -1.
- Iodine has been oxidised as the number of electrons has decreased as in its elemental form iodine has an oxidation state of 0.

B is incorrect as the oxidation state of the S in the SO_4^{2-} ion has not changed in NaHSO_4 but has decreased in H_2S .

C is incorrect as the oxidation state of the Na^+ ion has not changed.

D is incorrect as the oxidation state of the H^+ ion has not changed.

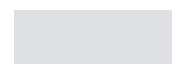
6

The correct answer is **D** because:

- Oxidation refers to the **loss of electrons** and reduction refers to the **gain of electrons**: OIL-RIG (oxidation is loss - reduction is gain).
- In the compound TiO_2 , oxidation has an oxidation state of -2.
- Oxygen has increased its oxidation number to 0 in O_2 as in the elemental form it has an oxidation state of 0.

A is incorrect as the oxidation state of fluorine has not changed.

B is incorrect as the oxidation number of bromine decreases in this reaction.



A is incorrect as in $\text{Mg}(\text{OH})_2$ the oxidation number of magnesium is +2, oxygen is -2 and hydrogen is +1.

B is incorrect as in Na_2SO_4 the oxidation number of sodium is +1, sulfur is +6 and oxygen is -2.

D is incorrect as in NH_4Cl the oxidation number of nitrogen is -3, hydrogen is +1 and chlorine is -1.

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The correct answer is **A** because:

- Oxidation refers to the **loss of electrons** and reduction refers to the **gain of electrons**: OIL-RIG (oxidation is loss - reduction is gain).
- In the compound NaI , sodium has an oxidation state of +1, and iodine an oxidation state of -1.
- Iodine has been oxidised as the number of electrons has decreased as in its elemental form iodine has an oxidation state of 0.

B is incorrect as the oxidation state of the S in the SO_4^{2-} ion has not changed in NaHSO_4 but has decreased in H_2S .

C is incorrect as the oxidation state of the Na^+ ion has not changed.

D is incorrect as the oxidation state of the H^+ ion has not changed.

6

The correct answer is **D** because:

- Oxidation refers to the **loss of electrons** and reduction refers to the **gain of electrons**: OIL-RIG (oxidation is loss - reduction is gain).
- In the compound TiO_2 , titanium has an oxidation state of +2.
- Oxygen has increased its oxidation number to 0 in O_2 as in the elemental form it has an oxidation state of 0.

A is incorrect as the oxidation state of fluorine has not changed.

B is incorrect as the oxidation number of bromine decreases in this reaction.

C is incorrect as the oxidation state of titanium has not changed.

7

The correct answer is **A** because:

- Oxidising agents are substances that oxidise other species, gain electrons and are themselves reduced.
- Write down the oxidation numbers of each species in the reaction.**



- In equation **A**, hydrogen oxidises Na(0) to Na⁺(+1) and is itself reduced from H(0) to H⁻(-1).

B, C & D are incorrect as hydrogen is acting as a reducing agent in all the other reactions shown; this means that it is losing electrons.

8

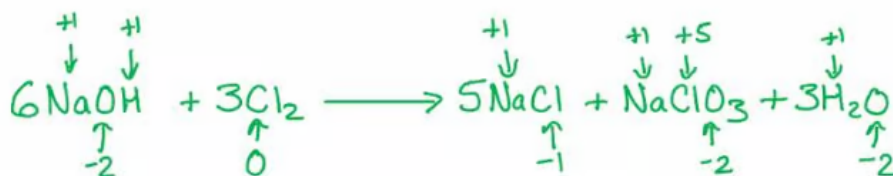
The correct answer is **A** because:

- The reaction between chlorine and cold sodium hydroxide solution with oxidation states:



- We can see that this is a **disproportionation reaction** in which chlorine is both oxidised and reduced.
 - Oxidation - change in oxidation number Cl: 0 to +1 in NaClO

- Reduction – change in oxidation number Cl: 0 to -1 in NaCl
- The reaction between chlorine and hot sodium hydroxide is also a **disproportionation reaction**:



- Oxidation – change in oxidation number Cl: 0 to +5 in NaClO₃
- Reduction – change in oxidation number Cl: 0 to -1 in NaCl
- Therefore the correct option is **A**.

B, C & D are incorrect as these answers do not correspond to the actual oxidation numbers of chlorine.

9

The correct answer is **C** because:

- In reaction 1 the bromide ions reduce the sulfuric acid to sulfur dioxide gas, this decreases the oxidation state of the sulfur from +6 in the sulfuric acid to +4 in the sulfur dioxide.
 - A change in the oxidation state of 2.
- In reaction 2 the reduction of sulfuric acid (oxidation state +6) is more complex.
 - The first stage is to sulfur dioxide (sulfur oxidation state +4).
 - Then to sulfur (oxidation state 0).
 - And then to hydrogen sulfide (sulfur oxidation state -2).
 - This makes the greatest change in oxidation state in this reaction 8 (from +6 to -2).
- In reaction 3 there is no change in the oxidation state of sulfur.
 - Both hydrogen and potassium have an oxidation state of +1 meaning that the sulfur oxidation state in these compounds is +6.

A, B & D are incorrect as these answers do not correspond to the actual changes in oxidation number involved.